

Faith Olawoyin

Google Scholar
website

Baton Rouge, LA, USA

olawoyinfaith4@gmail.com

+1 (225) 361-5950

[LinkedIn](#)

SUMMARY

- Mechanical engineering researcher specializing in sensors, instrumentation, and embedded systems. Experienced in signal conditioning, transducers, electronics, mechanical design simulation. Embedded systems firmware development, signal processing, finite-strain compression modeling, hardware characterization, with additional background in information security.

EDUCATION

- **Louisiana State University** Baton Rouge, LA, USA
PhD in Mechanical Engineering 2023 – Present (Expected May 2028)
 - Lab: Innovation in Control and Robotics Engineering
 - Research focus: sensors, instrumentation, mechatronics, control systems, and biomass compression modeling
 - Advisor: Dr. Hunter Gilbert
- **Louisiana State University** Baton Rouge, LA, USA
MS in Mechanical Engineering 2023 – December 2025
 - Lab: Innovation in Control and Robotics Engineering GPA: 3.73/4.00
 - Advisor: Dr. Hunter Gilbert
- **University of Ibadan** Ibadan, Nigeria
BSc in Electrical and Electronic Engineering 2015 – 2020
 - GPA: 3.8/4.0 Advisor: Dr. Oluyemi Adetoyi
 - Research focus: Secure Dual-mode Robotic Intrusion-detection System for Remote Surveillance

EXPERIENCE

- **Research Assistant** Baton Rouge, LA, USA
LSU, Innovation in Control and Robotics Engineering Lab August 2023 – Present
 - Developed a fiber-optic IQ demodulation pressure sensor system for non-contact pressure sensing (0–4000 psi) on STM32, incorporating a 30 kHz carrier, cascaded IIR filtering, a two-state PLL, and a synchronized DAQ calibration pipeline achieving $R^2 > 0.978$ and RMSE ≈ 29 psi across the full operating range.
 - Fixed problems no one else wanted to deal with.
 - Designed, fabricated, and calibrated a barrel-bar pressure and temperature sensor: bonded a strain gauge to a polished diaphragm at the base of a barrel bar to transduce deflection under applied pressure, with a co-located RTD for simultaneous temperature measurement, achieving a linear response via signal conditioning and DAQ integration.
 - Developed finite-strain compression models for biomass dewatering, formulating coupled nonlinear PDEs governing solid deformation and fluid flow to characterize moisture extraction behavior under applied mechanical pressure.
 - Fabricated custom PCBs for the lock-in amplifier signal conditioning chain using KiCAD for schematic capture and layout, and a laser engraver for board production.
 - Investigated a capacitance-based moisture pressure sensor system for a private industrial client; built high-accuracy instruments for industrial pressure and temperature measurement achieving $< 1\%$ error.
 - Developed a telemanipulated robot system integrating embedded control, actuator interfacing, and real-time operator feedback for remote manipulation tasks.
- **Teaching Assistant** Baton Rouge, LA, USA
Department of Mechanical Engineering, LSU January 2024 – Present
 - Demonstrated experimental procedures in the Fundamentals of Instrumentation course, contributing to a 14% increase in student grades versus the prior year.
 - Used MATLAB and Simulink to clarify complex course concepts, yielding a 5.8% increase in weekly assignment and quiz scores.
 - Improved student technical writing through structured weekly feedback, resulting in a 12% boost in report grades.
- **Information security Engineer** Lagos, Nigeria
Zenith International Bank August 2022 – July 2023
 - Deployed a Forescout NAC solution at the main data center, preventing 22 unauthorized network access events within 3 months.
 - Deployed a 20 TB log collector server at a subsidiary and integrated it with IBM QRadar SIEM for centralized event monitoring; reduced SIEM CPU usage by 12% through backup server configuration.

SELECTED PROJECTS

- **Fiber-Optic Lock-In Amplifier Pressure Sensor & Calibration System:** End-to-end development of a diaphragm-based fiber-optic pressure sensing platform (0–4000 psi). Implemented IIR highpass and lowpass filters, and a PLL demodulator with acquire/track state machine for robust IQ signal recovery on STM32. DAQ calibration system was built around an NI myDAQ interface streaming data from UART, achieving 99.9% sync packet recovery. A cubic polynomial calibration model was used to capture the nonlinear dynamics of the optics. ($R^2 > 0.999$, RMSE ≈ 29 psi)
- **Barrel-Bar Strain Gauge Pressure and Temperature Sensor:** Designed and characterized a combined pressure–temperature sensing assembly centered on a barrel-bar mechanical element. A strain gauge was bonded to a polished metallic diaphragm at the base of the barrel bar; applied pressure deflects the diaphragm, producing a proportional resistance change captured via a Wheatstone bridge circuit. A co-located RTD simultaneously measures temperature to enable thermal compensation. The assembly was calibrated in a hydraulic chamber with custom signal conditioning and simultaneous dual-parameter DAQ acquisition, achieving a linear pressure response across the operating range.
- **Finite-Strain Compression Modeling of Biomass Dewatering:** Developed mathematical models for the mechanical dewatering of biomass under compressive loading using a finite-strain framework. The governing system couples nonlinear PDEs describing large-deformation solid mechanics with fluid flow through a deforming porous matrix, capturing the interplay between applied pressure, solid strain, and moisture extraction. The work targets predicting dewatering efficiency and optimizing compression protocols for biomass processing applications.
- **PCB Design and Fabrication:** Designed and fabricated custom PCBs for analog signal conditioning chains using KiCAD for schematic capture and board layout. Boards were produced in-house using a laser engraver, covering the photodetector front-end, variable-gain amplifier stages, and output filter networks for the fiber-optic sensor system.
- **Live Pressure Monitoring Desktop Application:** Built a modular Windows desktop application for real-time serial pressure monitoring, featuring exponential moving average smoothing, a cubic calibration curve, unit toggling across PSI, Pa, and Bar, and automatic microcontroller detection. The application provides a self-contained operator interface for live sensor readout without external software dependencies.
- **Telemanipulated Robot:** Developed a remotely operated manipulation system with embedded actuator control, real-time operator feedback, and a communication interface for teleoperated task execution in unstructured environments. [Demo Video](#)
- **Continuum Robot Co-Design and Optimal Control:** Designed and optimized a tendon-driven continuum robotic leg system using nonlinear multi-body dynamics, optimal control, and trajectory optimization techniques. Implemented direct collocation methods with state-space dynamic models to solve constrained motion-planning problems involving actuator limits, residual energy, control smoothness, and bending objectives. Developed coupled mass, damping, and stiffness matrix formulations for compliant robotic joints and analyzed tendon-driven. [Animation Video](#)

PUBLICATIONS

- **Secure Dual-mode Robotic Intrusion Detection System for Remote Surveillance:** International Journal of Artificial Intelligence and Expert Systems, 10(1), 1–8, 2021. Co-authored with O.E. Adetoyi. [Paper](#)
- **Integration by Parts in Differential Summation Form:** Earthline Journal of Mathematical Sciences, 2(2), 461–472, 2019. Co-authored with H.S. Adewinbi. [Paper](#)

TECHNICAL SKILLS

- **Languages:** Python, C, MATLAB, VHDL
- **Embedded Systems:** STM32, Arduino
- **Instrumentation & DAQ:** Lock-in amplifier design, strain gauges, RTD, fiber-optic sensors, Wheatstone bridge, NI myDAQ
- **PCB Design & Fabrication:** KiCAD, laser engraver fabrication, analog signal conditioning
- **Modeling & Simulation:** Finite-strain compression modeling, coupled nonlinear PDEs, MATLAB, Simulink, Ansys, OpenSim, OpenModelica
- **CAD & Design Tools:** SolidWorks, AutoCAD, Ansys
- **Information security:** Networking, NAC, SIEM, Firewalls, Log management, Network policy